MS Digital Biomarkers

Research Grade

Clinical metrics extracted from 6 sessions · Last 30 days

FATIGUE INDEX

0.81

None — Severe

High

>0.5 = severe fatigue · may indicate relapse or poor disease control

COEFFICIENT OF VARIATION (CV)

0.037

Consistent — Erratic

Normal consistency

CV >0.35 = clinically significant RT variability · IIV MS biomarker

RELIABLE CHANGE INDEX (RCI)

+1.86

Declining — Improving

Stable — no significant change

|RCI| >1.96 = statistically significant change (Jacobson & Truax 1991)

EWMA TREND ANALYSIS

Improving · 0.449

Current EWMA: 56.3

REACTION TIME TREND

Declining · 0.288

Current EWMA: 1182 ms

Contextual Correlations (Pearson r — how much each factor affects performance)

Fatigue Level

Score impact	r = 0.18
RT impact	r = 0.42

Weak positive effect on score · n=6

Sleep Quality

Score impact	r = 0.86
RT impact	r = -0.90

Strong positive effect on score · n=6

Medication Timing

Score impact	r = -0.42
RT impact	r = 0.20

Moderate negative effect on score · n=6

● Normal / Improving ● Moderate / Watch ● High / Declining | *Based on BICAMS standards · RCI threshold ±1.96*